

Model architecture and pipeline design

From raw farmer input — text, photo, or voice — to delivered advice, inside a 200–300 ms budget on the core reasoning path.

Scope of this milestone

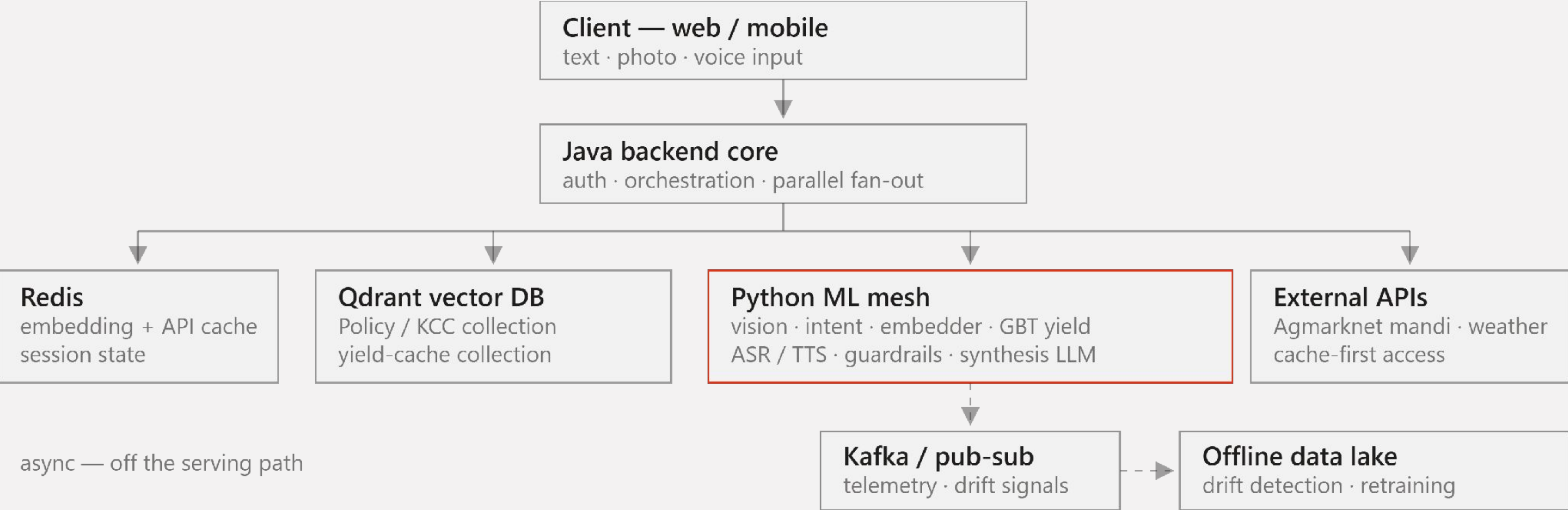
Selects and justifies the model behind each module — vision, retrieval, yield, synthesis, voice.

Decide the overall architecture of the project.

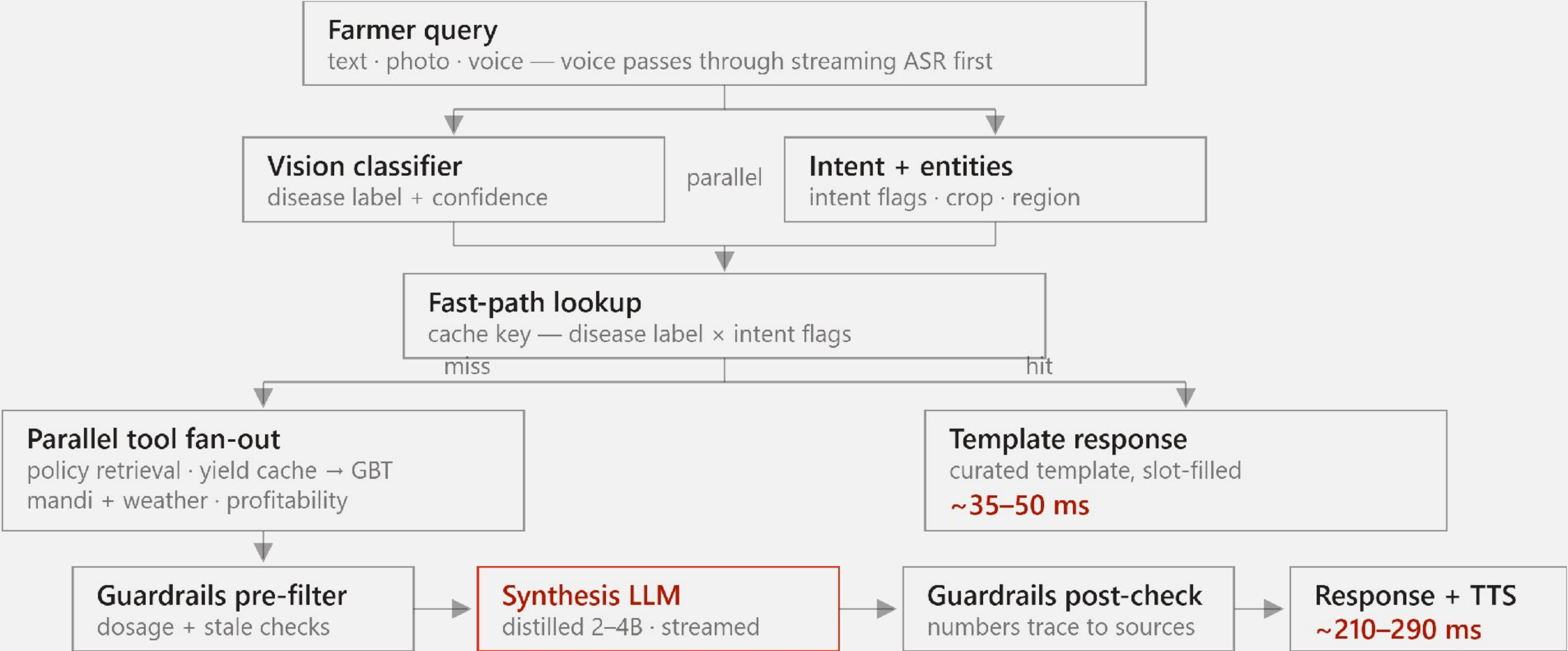
Justification of the architecture, tools and models used.

Design-phase status: latency figures and hyperparameters are estimates until Milestone 4 measures them.

System architecture



Request flow, input to answer



Model selection by module

MODULE	MODEL	ORIGIN	SIZE
Vision — disease	ViT-Small (edge fallback: MobileViT-XXS)	ImageNet-pretrained → fine-tuned	22–86M
Text embedding	MuRIL-base — code-mix aware	pretrained · optional contrastive tune	236M
Intent + entities	DistilBERT-class, joint head	pretrained → fine-tuned	66M
Yield prediction	XGBoost / LightGBM	trained from scratch	tree ensemble
Synthesis	Gemma — 12B teacher, distilled student	distilled in Milestone 4	2–4B
ASR	IndicWhisper / Bhashini-class	pretrained → dialect fine-tune	0.25–1.5B
TTS	VITS / FastSpeech2-class, Indic	pretrained	30–90M

Design decisions and alternatives

Distill 12B → 2–4B for synthesis

A 12B decoder cannot generate inside the 150–180 ms slice. The student keeps the teacher’s domain behavior at a servable size.

Tool selection before generation

The extractor picks tools upfront; all fire in one parallel batch. A turn-by-turn agent loop survives only as the slow-path fallback.

GBT for yield; vectors only as cache

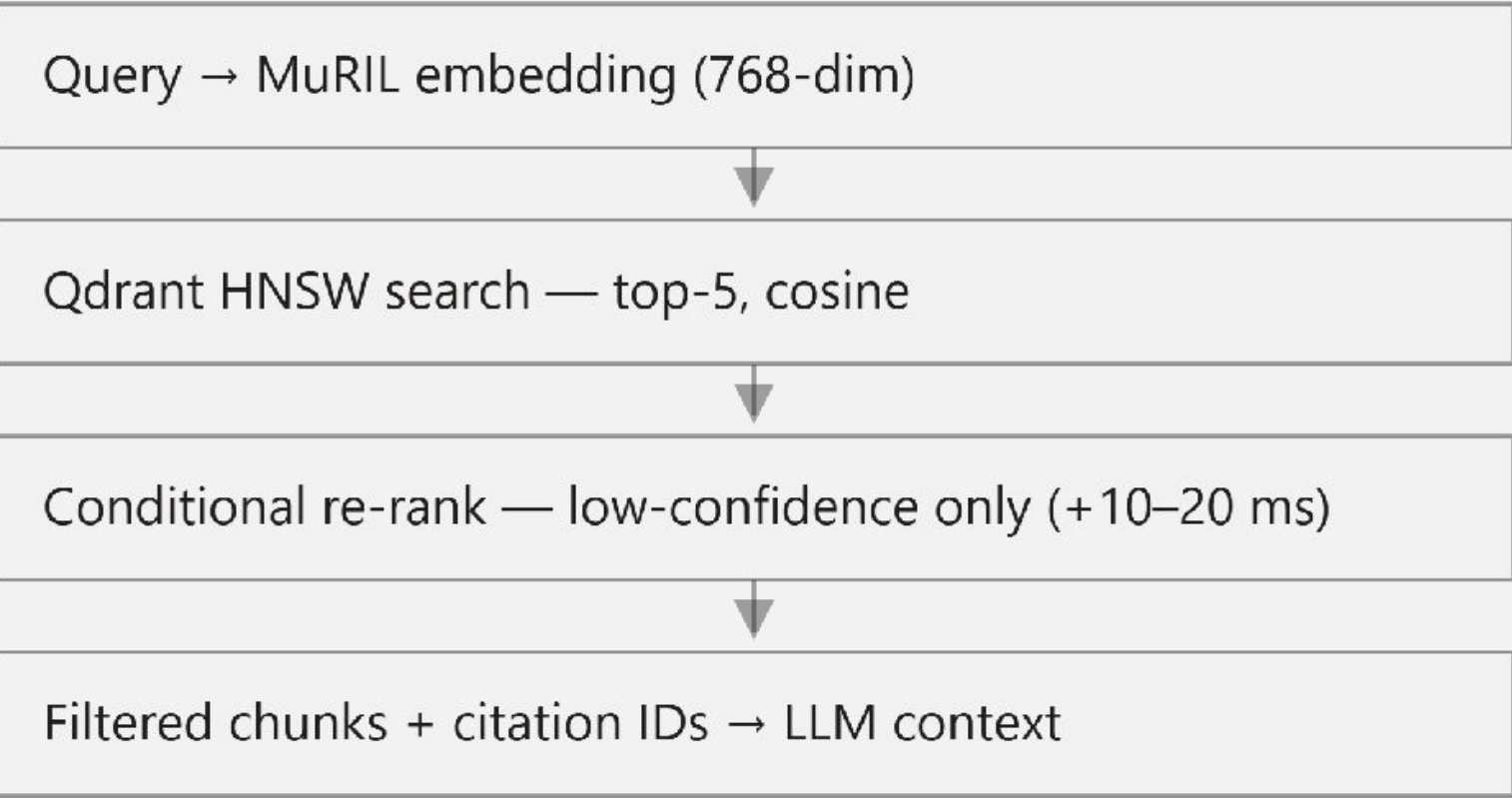
k-NN averaging underperforms boosted trees on tabular regression — the cache sits in front of the model, never in place of it.

Rule-based guardrails, two stages

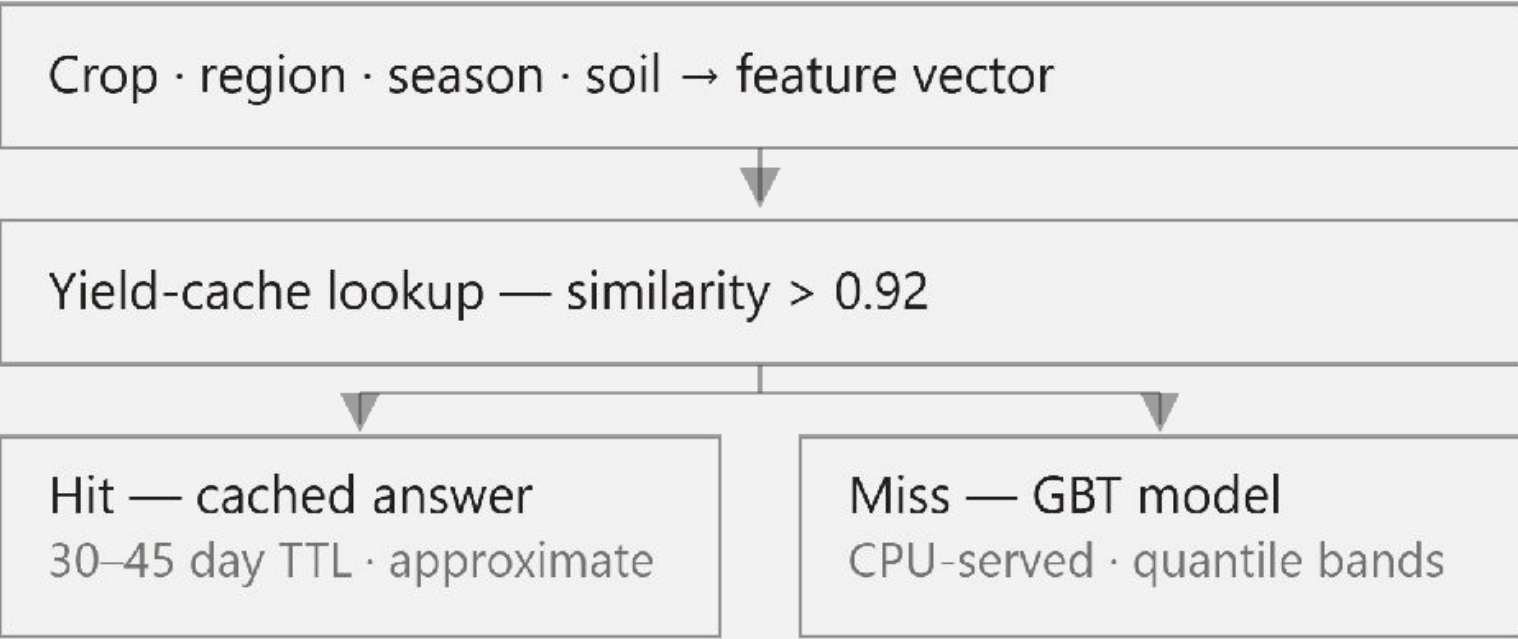
Pre-filter on retrieved data, post-check on generated text — no second generation pass, and every rule stays auditable.

Retrieval and yield caching

Policy / KCC retrieval



Yield estimate



Two separate embedding spaces — the semantic policy index never shares a collection with the structured yield cache.

Synthesis and guardrails

Prompt assembly — fixed order

01	System rules	role, safety limits, format
02	Retrieved chunks	policy passages + citation IDs
03	Tool data	yield · price · profitability · flags
04	Farmer query	original text, post-ASR if spoken
05	Output format	guided decoding, 60–100 tokens

Two-stage guardrails

Pre-filter	validates retrieved data before synthesis — dosage bounds, staleness flags
Post-check	scans generated text — untraced numbers, banned terms, language
Fail-closed	a guardrail outage never returns unvalidated advice
No agent loop	tools fire upfront in parallel; a dynamic loop only as slow-path fallback

Latency budget and serving

design estimates — benchmarked in Milestone 4

PATH	ESTIMATE	WHAT IT COVERS
Fast path	~35–50 ms	template served on a cache hit
Slow path	~210–290 ms	full fan-out, synthesis, both guardrail stages
Voice input	+150–400 ms	ASR finalization before the pipeline starts
Voice output	streamed	TTS begins on the first completed sentence

Synthesis LLM

A100/H100-class, FP8 on vLLM —
batched, with speculative decoding.

Small models

T4/L4-class or CPU — TensorRT / ONNX
for vision, intent, embedders.

Yield model

CPU only — the boosted-tree ensemble
needs no GPU anywhere in its path.

Validation plan for Milestone 4

Assumptions to validate — Appendix G

150–180 ms generation slice — throughput benchmark on the distilled checkpoint

Student accuracy vs the 12B teacher — held-out agri-QA set

GBT vs k-NN and neural baselines — cross-validated on real yield data

Guardrail recall — adversarial red-team queries

MuRIL vs multilingual-e5 — recall@k on code-mixed queries

Milestone 4 activities

Assemble and clean the training datasets

Fine-tune the vision classifier; per-class F1 baseline

Build the distillation pipeline — teacher output → student fine-tune

Curate the initial fast-path template set

Integrate and load-test against the latency targets